# One-to-One and Onto Linear Transformations

## 1. What is a Linear Transformation?

A linear transformation $T : V \to W$ is a function between two vector spaces $V$ and $W$ that preserves the operations of vector addition and scalar multiplication. That is, for any vectors $\mathbf{u}, \mathbf{v} \in V$ and any scalar $c$, the following properties must hold:

1. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$
2. $T(c\mathbf{u}) = cT(\mathbf{u})$

## 2. One-to-One (Injective) Linear Transformations

A linear transformation $T : V \to W$ is **one-to-one** if every distinct vector in $V$ maps to a distinct vector in $W$. In other words, no two different vectors from the domain $V$ can map to the same vector in $W$ (no repeated outputs).

More formally, $T$ is one-to-one if:

$$T(\mathbf{v}_1) = T(\mathbf{v}_2) \text{ then } \mathbf{v}_1 = \mathbf{v}_2$$

Alternatively, a transformation is one-to-one if the **kernel** of $T$ only contains the zero vector. The kernel of $T$ is the set of all vectors $\mathbf{v} \in V$ such that $T(\mathbf{v}) = \mathbf{0}$. Thus:

$$\ker(T) = \{\mathbf{0}\}$$

We represent linear transformations as matrices, thus a linear transform is one-to-one if $T(\mathbf{v}) = A\mathbf{v} = \mathbf{0}$ has the trivial solution $\mathbf{v} = \mathbf{0}$, which means that the column vectors of $A$ form a linearly independent set. So, if the transformation is represented by a matrix $A$, the **transformation is one-to-one if the columns of $A$ are linearly independent.**

**If the matrix has more columns than rows, the transformation cannot be injective** because there will necessarily be more variables than equations, leading to free variables and multiple solutions. This would mean that some distinct input vectors could map to the same output vector (violating the injective condition).

## 3. Onto (Surjective) Linear Transformations

A linear transformation $T : V \to W$ is **onto** if for every vector $\mathbf{w} \in W$, there is at least one vector $\mathbf{v} \in V$ such that $T(\mathbf{v}) = \mathbf{w}$. In other words, the transformation covers the entire codomain $W$. (Every vector in $W$ has something from $V$ that maps to it. The transformation must "touch" every vector in $W$.)

More formally, $T$ is onto if:

$$\forall\, \mathbf{w} \in W,\ \exists\, \mathbf{v} \in V \text{ such that } T(\mathbf{v}) = \mathbf{w}$$

This means the **image** (or range) of $T$, which is the set of all vectors $T(\mathbf{v})$ for $\mathbf{v} \in V$, is equal to the codomain $W$:

$$\mathrm{Im}(T) = W$$

We represent linear transformations as matrices, thus a linear transform is onto if $T(\mathbf{v}) = A\mathbf{v} = \mathbf{b}$ has a solution for every $\mathbf{b}$ in $W$, which means that the **span of the column vectors of $A$ must be equal to the whole space $W$.**

**If the matrix has more rows than columns, the transformation cannot be surjective** because the image (span of the columns) would be confined to a lower-dimensional subspace of the codomain. This would leave parts of the codomain uncovered, meaning there would be vectors in the codomain that do not have a pre-image in the domain (violating the surjective condition).

## 4. One-to-One and Onto (Bijective) Transformations

A transformation that is both one-to-one and onto is called **bijective**. A bijective transformation is a perfect pairing between the domain and codomain, where every vector in the domain maps to a unique vector in the codomain, and every vector in the codomain has a corresponding pre-image in the domain.

A bijective transformation $T$ has an inverse transformation $T^{-1}$ that maps vectors from $W$ back to $V$. In matrix form, if $T(\mathbf{v}) = A\mathbf{v}$, the matrix $A$ is invertible if and only if $T$ is bijective.

A linear transformation is bijective if it is both one-to-one and onto. For the matrix representing the transformation to satisfy both these conditions the matrix must have as many columns as rows, ensuring that it can be both injective and surjective. In other words, **the matrix must be square** (i.e., the number of rows = the number of columns).

## 5. Examples

**Example 1:** If $T : \mathbb{R}^2 \to \mathbb{R}^2$ is defined by $T(\mathbf{v}) = A\mathbf{v}$, where $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$, then $T$ is one-to-one because the columns of $A$ are linearly independent, and the kernel only contains the zero vector. $T$ is also onto as the span of the column vectors will give the entire set, $\mathbb{R}^2$. (A square matrix with linearly independent column vectors represents a bijective linear transformation. )

**Example 2:** If $T : \mathbb{R}^3 \to \mathbb{R}^2$ is defined by $T(\mathbf{v}) = A\mathbf{v}$, where $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$, then $T$ is onto because the two column vectors $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ of $A$ span $\mathbb{R}^2$. This ensures that every vector in $\mathbb{R}^2$ can be expressed as a linear combination of the rows of $A$. However, this transformation is NOT one-to-one as the column vectors of $A$ do not form a linearly independent set. The column vector $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is a linear combination of the first two column vectors. As such, the kernel of the transform is the set of all vectors in $\mathbb{R}^3$ of the form $= \begin{bmatrix} -t \\ -t \\ t \end{bmatrix}$ where $t$ is a real number.

*Notice that the number of rows of A are less than the number of columns of A, thus this transform can not be one-to-one.*

**Example 3:** If $T : \mathbb{R}^3 \to \mathbb{R}^4$ is defined by $T(\mathbf{v}) = A\mathbf{v}$, where $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$, then $T$ is one-to-one as the columns of $A$ are clearly linearly independent. However, $T$ is NOT onto as the image of $T$ is spanned by the first 3 columns, which only spans a 3-dimensional subspace of $\mathbb{R}^4$ (which is not all of $\mathbb{R}^4$).

*Notice that the number of columns of A is less than the number of rows of A, thus this transform can not be onto.*